

A.N.B. Lactated Ringer's Inj

Controlled Medicine/*Ubat Terkawal*

Each 100 mL contains:	
Sodium Chloride	600 mg
Lactic Acid and Sodium Hydroxide	
Equivalent to Sodium Lactate	310 mg
Potassium Chloride	30 mg
Calcium Chloride	20 mg
Sterile Water for Injection ad to	100 mL

Product Description: Sterile, clear, colourless, pyrogen-free solution.

Pharmacodynamics/ Pharmacokinetics: Time to peak effect: 8 to 12 hours.

Indication: Fluid and electrolyte replenisher.

Recommended Dosage: Intravenous infusion or as physician's order.

Contraindications: Patients with lactic acidosis.

Warnings and Precautions: None

Drug Interactions: Cautiously administer parenteral fluids, especially those containing sodium ions, to patients receiving corticosteroids or corticotropin. There is a hazard in prescribing potassium in digitalis intoxication manifested by atrioventricular conduction disturbance. In digitalized patients with severe or complete heart block, do not use potassium salts. Administer calcium very cautiously to a digitalized patient: the combination may precipitate arrhythmias.

Adverse Effects:

Calcium: Calcium salts are irritating to tissue when administered by IM or subcutaneous injection and cause mild to severe local reactions including burning, necrosis and sloughing of tissue, cellulitis, and soft tissue calcification; venous irritation may occur with IV administration. When injected IV, calcium salts should be administered slowly through a small needle into a large vein to avoid too rapid an increase in serum calcium and extravasation of calcium solution into the surrounding tissue with resultant necrosis. Patients may complain of tingling sensations, a sense of oppression or heat waves, and a calcium or chalky taste following IV administration of calcium salts. Rapid IV injection of calcium salts may cause vasodilation, decreased blood pressure, bradycardia, cardiac arrhythmias, syncope, and cardiac arrest. When administered intracardially, calcium should be injected directly into the ventricular cavity. If the drug is injected into the myocardium, coronary arteries may be lacerated, cardiac tamponade or pneumothorax may be produced, and intractable ventricular fibrillation may result. Calcium salts may cause constipation. Calcium chloride, by any route of administration, produces more irritation than the other calcium salts. Hypercalcemia is rarely produced by administration of calcium alone, but may occur when large doses are given to patients with chronic renal failure. Since hypercalcemia may be more dangerous than hypocalcemia, overtreatment of hypocalcemia should be avoided.

Potassium: Hyperkalemia: Adverse reactions involve the possibility of potassium intoxication. Signs and symptoms include; Paresthesias of extremities; flaccid paralysis; muscle or respiratory paralysis; areflexia; weakness; listlessness; mental confusion; weakness and heaviness of legs; hypotension; cardiac arrhythmias; heart block; ECG abnormalities such as disappearance of P waves, spreading and slurring of the QRS complex with development of a biphasic curve and cardiac arrest.

GI and other local effects: Adverse effects of potassium salts may include nausea, vomiting, diarrhea, flatulence, and abdominal pain or discomfort. Pain at the site of injection and phlebitis may occur during IV administration of solution containing 30 mEq or more potassium per liter.

Reactions due to solution or technique of administration: Febrile response; infection at injection site; venous thrombosis; phlebitis extending from injection site; extravasation; hypervolemia; hyperkalemia; venospasm. Hypermnatremia may be associated with edema and exacerbation of CHF due to retention of water, resulting in expanded extracellular fluid volume.

Ion excess/deficit: Symptoms may result from an excess or deficit of one or more of the ions present in the solution; therefore, frequent monitoring of electrolyte levels is essential. If infused in large amounts, chloride ions may cause a loss of bicarbonate ions, resulting in an acidifying effect.

Postoperative salt intolerance: Symptoms include cellular dehydration; weakness; disorientation; anorexia; nausea; distention; deep respiration; oliguria; increased BUN.

Too rapid infusion of hypertonic solution may cause local pain and venous irritation. Adjust rate of administration according to tolerance. Use of the largest peripheral vein and a well-placed small bore needle is recommended. If an adverse reaction occurs, discontinue the infusion, evaluate the patient, institute appropriate countermeasures and save the remainder of the fluid for examination.

Treatment of Overdose: None

Storage Conditions: Store below 30°C. Keep out of reach of children. Discard unused portion.

Packaging: Polypropylene bottle of 20 x 500 mL.

Polypropylene bottle of 10 x 1000 mL.

Manufacturer: A.N.B. Laboratories Co., Ltd.

557 Ramintra Rd., Bangkok, Thailand.

Product Registration Holder: Delfi Marketing Sdn. Bhd. (15872-P)

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